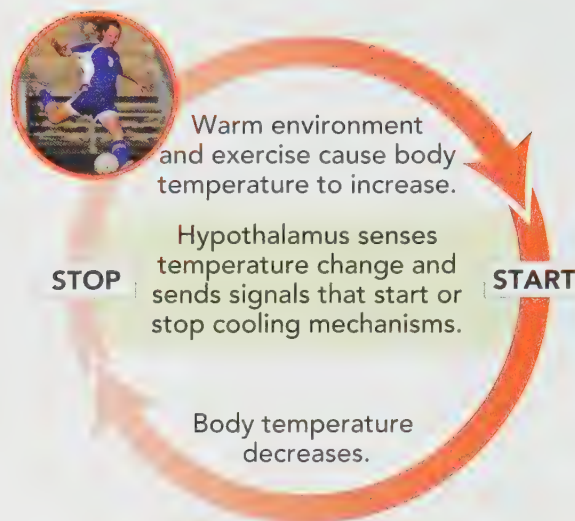
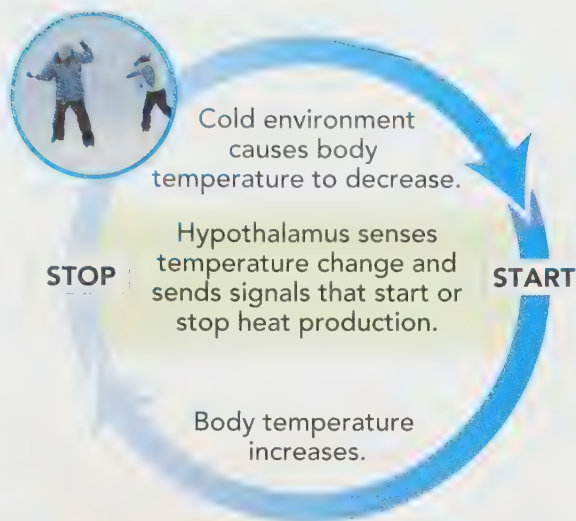




INTERACTIVITY

Figure 27-4

Body Temperature Control

In the human body, temperature is controlled through various feedback inhibition mechanisms.

A Living Example Could biological systems achieve homeostasis through feedback inhibition? Absolutely. One example is the maintenance of body temperature. **Figure 27-4** shows how the body regulates temperature in a way similar to that of a home heating system.

A part of the brain called the hypothalamus contains nerve cells that monitor body temperature. If the nerve cells sense that the core temperature has dropped much below 37°C , the hypothalamus produces chemicals that signal cells throughout the body to speed up their activities. Heat produced by this increase in activity, especially cellular respiration, causes a rise in body temperature, which is detected by nerve cells in the hypothalamus.

Have you ever been so cold that you began to shiver? If your body temperature drops well below its normal range, the hypothalamus releases chemicals that signal muscles just below the surface of the skin to contract involuntarily—to “shiver.” These muscle contractions release heat, which helps the body temperature to rise.

If body temperature rises too far above 37°C , the hypothalamus slows down cellular activities to reduce heat production. This is one of the reasons you may feel tired and sluggish on a hot day. The body also responds to high temperatures by producing sweat, which helps to cool the body surface by evaporation.

HS-LS1-2

Quick Lab

Guided Inquiry

How Do You Respond to an External Stimulus?



1. Have your partner put on safety goggles.
2. Crumple up a sheet of scrap paper into a ball.
3. Watch your partner's eyes carefully as you toss the paper ball toward his or her face.
4. Repeat step 3, three times.
5. Exchange roles and repeat steps 1, 3, and 4.

ANALYZE AND CONCLUDE

1. **Observe** Describe your partner's reaction to step 3.
2. **Compare and Contrast** Did you see any change in behavior as you repeated step 3? Explain.
3. **Infer** What is the function of the blink reflex?
4. **Develop Models** Draw a model to show how different systems of the body interacted.